Descriptive Analytics Report: Flipkart

Daily Purchase Behavior at Scale

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Section: BCADS24

Tracking Purchase Trends, Product Categories, Success Rates, Regional Demand & Operational Status

Note on data: This report uses a synthetically generated Flipkart-style dataset containing 200 records across 5 dates, 8 states and 6 purchase categories. The dataset is already aggregated rather than raw event-level order data. Therefore, the analysis describes the generated observations and does not represent real-time Flipkart order data.

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1. Objective

Build a descriptive analytics layer over Flipkart order data to answer the following recurring business questions:

- How are daily order volumes trending over time?
- How is daily GMV trending over time?
- Which purchase category contributes the highest order volume?
- Which purchase categories contribute the most to overall GMV?
- Which states generate the highest order volumes?
- Which states contribute the most GMV?
- Successful order level.
- State-level order & regional demand geographically.
- How does active seller activity relate to order volume?
- Weekday and weekend order patterns.
- Customer Satisfaction (CSAT).

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2. How This Would Work at Real Scale

Processing large-scale e-commerce order events is an engineering problem before it is an analytics problem. A realistic architecture would collect order events, aggregate them by time/category/state, store raw and aggregated data, and expose the resulting KPIs through a data visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the order dataset	Google, Claude & ChatGPT
Stream processing	Aggregate by category, state and time	Jupyter Notebook (Python)
Storage	Retain raw and processed data	Excel files / CSV files
Batch aggregation	Daily KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Daily Order Volume

Over the sampled 5-day window, the dataset records 1,35,30,622 orders and approximately ₹2,433.65 crore in GMV. The daily charts below show how order volume and GMV move across the available dates.

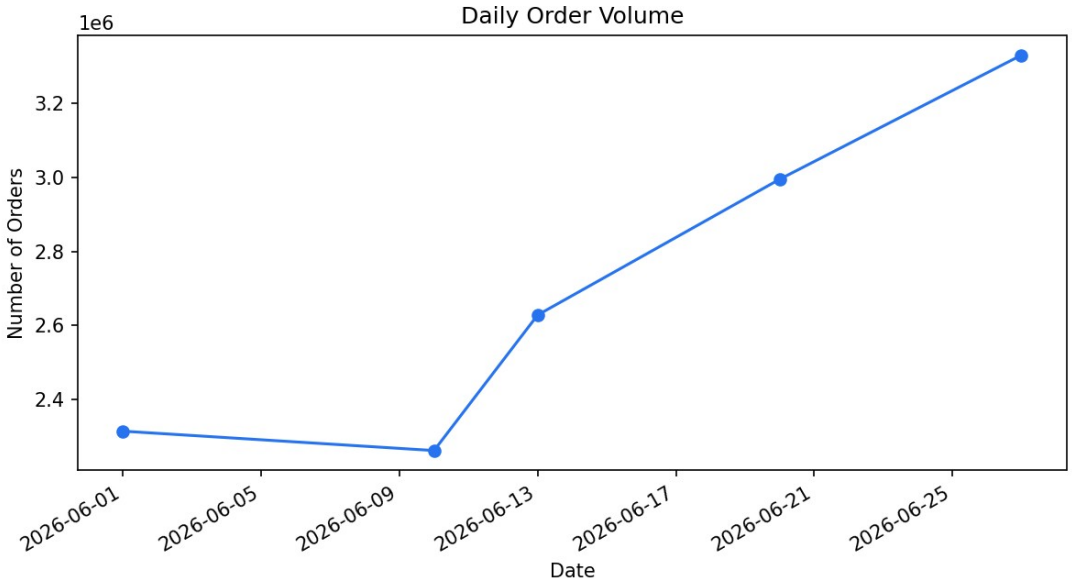


Figure 1. Daily order volume

INSIGHTS:

- Higher-volume dates indicate periods of stronger purchase activity.
- Lower-volume dates indicate comparatively weaker activity.
- Monitoring daily order volume can help identify demand fluctuations.
- This can support operational and logistics capacity planning.

KPI:

Total orders: 13.53 Million

4. Daily GMV Trending Over Time

Total GMV in the generated records is ~₹24.34 Billion. The average GMV per represented date is ~₹4.867 Billion, while the highest daily aggregate GMV is ~₹5.959 Billion. GMV is calculated directly from the dataset's GMV field.

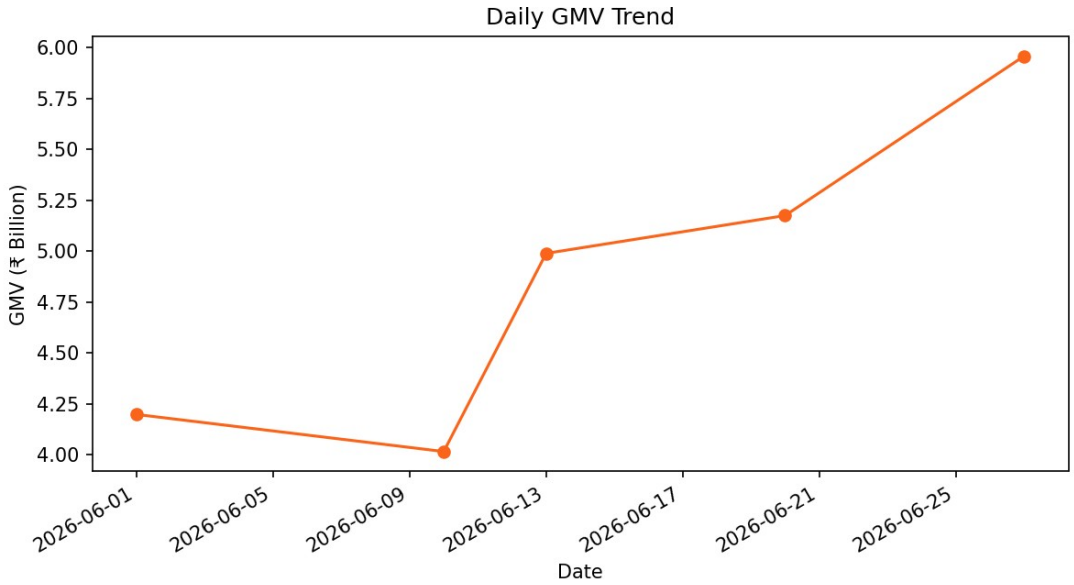


Figure 2. Daily GMV trend

INSIGHTS:

- GMV varies across the available dates.
- Higher GMV indicates stronger monetary purchase activity.
- Daily GMV monitoring helps identify high-value purchase periods.
- GMV should be considered alongside order volume to understand demand more completely.

KPI:

Total GMV: ₹24.34 Billion

Average GMV per represented date: ≈ ₹4.867 Billion

Peak daily GMV: ≈ ₹5.959 Billion

5. Best-Performing Purchase Categories

Across the generated dataset, Electronics is the leading purchase category by order volume, accounting for ≈30.6% of all orders. Unlike a single-category dataset, this sample spans 6 categories, so the comparison below reflects relative standing across all of them.

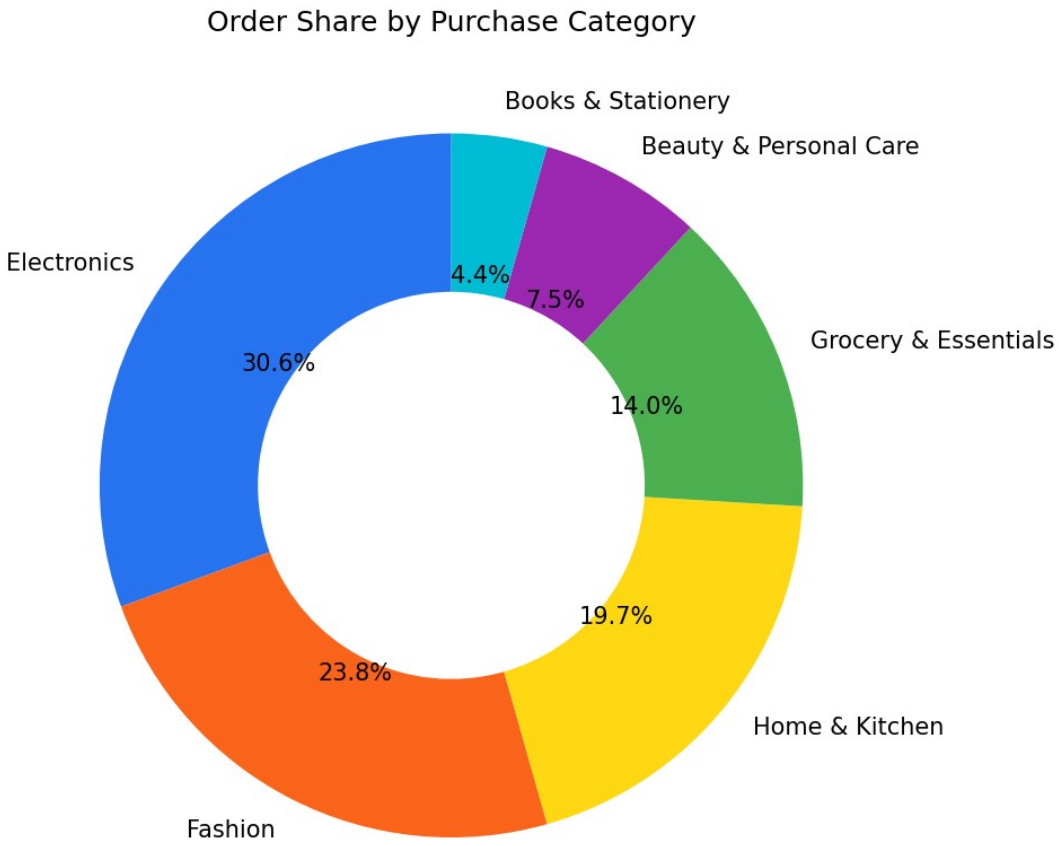


Figure 3. Order share by purchase category

INSIGHTS:

- Electronics leads order volume, followed by the next-largest categories in the chart.
- The distribution shows demand is spread unevenly across categories, with high-consideration categories like electronics and fashion dominating.
- This finding applies only to the generated dataset and the sampling assumptions used to build it.

KPI:

Top purchase category: Electronics

Order share: ≈30.6%

Categories represented: 6

6. Purchase Categories Contributing the Most to Overall GMV

Electronics is also the largest contributor to GMV, accounting for ≈39.9% of total GMV in the dataset — a higher share than its order-volume share, reflecting a higher average order value in this category.

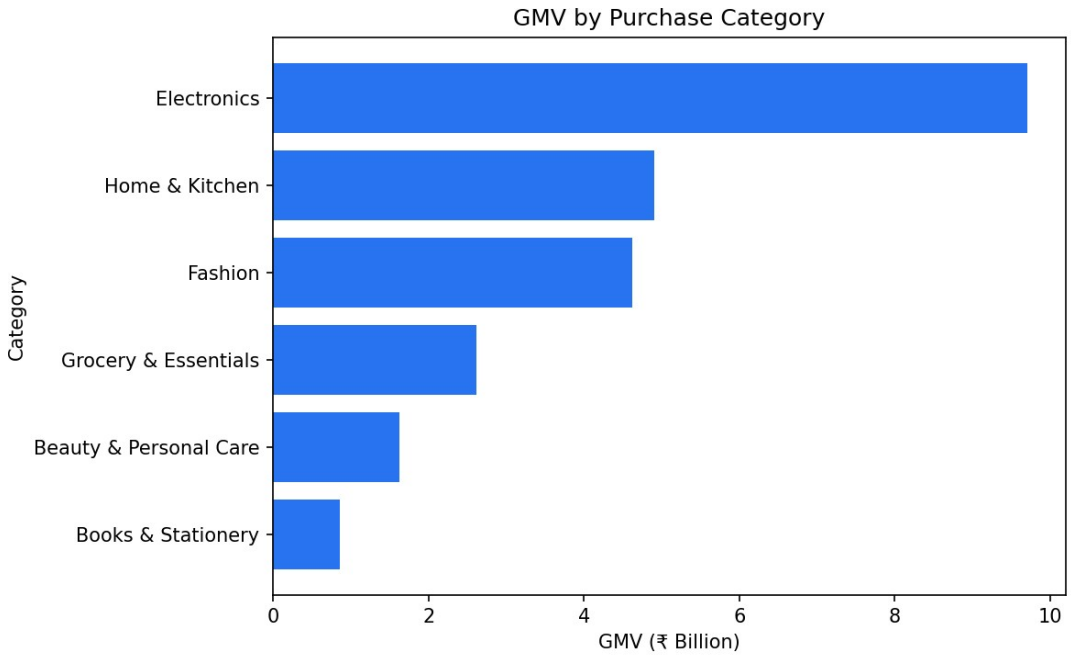


Figure 4. GMV by purchase category

INSIGHTS:

- GMV concentration by category does not always match order-volume concentration, since average order values differ by category.
- High-ticket categories such as electronics can contribute disproportionately more GMV than orders.
- The finding should be interpreted only within the generated dataset.

KPI:

Top GMV category: Electronics

GMV share: ≈39.9%

Categories represented: 6

7. Category Deep-Dive

The table below combines order volume, GMV, average order value, success rate, and CSAT for every purchase category, giving a single reference view of category-level performance.

Category	Orders	GMV (₹ Cr)	Avg Order Value (₹)	Success Rate	CSAT
Electronics	41,39,923	971.02	2,345	96.8%	4.57/5
Fashion	32,19,682	462.02	1,435	96.75%	4.5/5
Home & Kitchen	26,60,662	490.97	1,845	96.96%	4.53/5
Grocery & Essentials	18,98,858	261.77	1,379	97.06%	4.54/5
Beauty & Personal Care	10,16,660	162.27	1,596	97.02%	4.54/5
Books & Stationery	5,94,837	85.61	1,439	97.06%	4.59/5

INSIGHTS:

- Electronics has the highest average order value among the six categories, which explains why its GMV share (39.9%) exceeds its order-volume share (30.6%).
- Grocery & Essentials and Books & Stationery have the lowest average order values, consistent with typically low-ticket, high-frequency purchases.
- Success rate and CSAT are fairly stable across categories (roughly 96.7–97.1% and 4.5–4.6 respectively), suggesting operational reliability does not vary much by category in this sample.
- This view is useful for prioritizing which categories deserve deeper investment in catalog quality, logistics, or seller onboarding.

8. States Generating the Highest Order Volumes

Maharashtra is the largest state-level contributor by order volume, with 44,02,171 orders (32.53% of all sampled orders). Maharashtra is also the largest contributor by GMV, at ≈₹7.97 Billion. The distribution shows that demand is concentrated in a subset of states rather than evenly distributed.

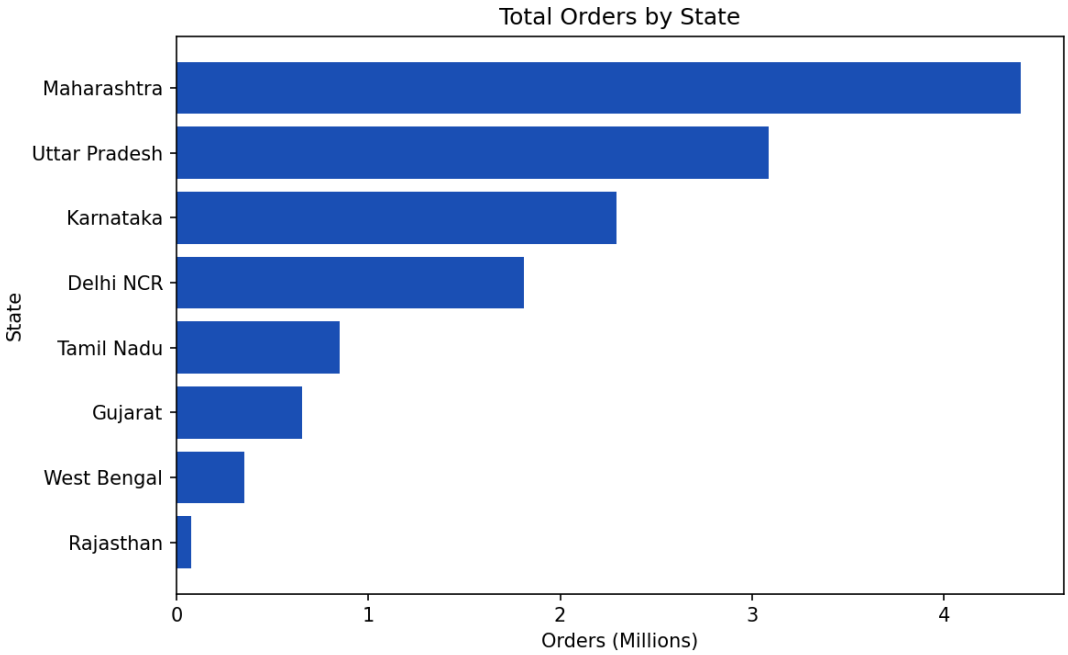


Figure 5. Total orders by state

INSIGHTS:

- Order demand is concentrated in a subset of states.
- Demand is not evenly distributed across all states.
- High-volume states may require greater fulfilment and logistics support.
- State-level order analysis can help with regional business planning.

KPI:

Top-performing state: Maharashtra

Recorded orders: 4.40 Million

Share of total orders: ≈32.53%

9. States Contributing the Most GMV

Maharashtra is also the largest contributor by GMV, at ≈₹7.97 Billion. The distribution shows that GMV, like order volume, is concentrated in a subset of states rather than evenly distributed.

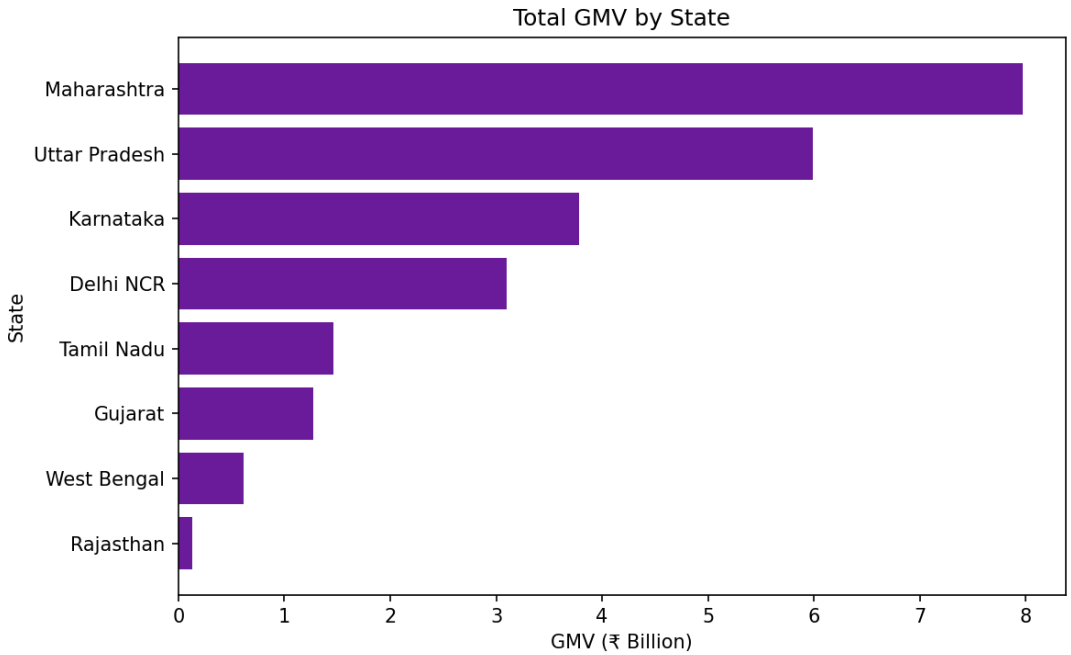


Figure 6. Total GMV by state

INSIGHTS:

- This indicates strong monetary purchase activity in the leading state.
- GMV is concentrated in a subset of states.
- High-GMV states can be important markets for operational and service planning.
- State-level GMV analysis can help identify commercially important regions.

KPI:

Top-performing state: Maharashtra

Recorded GMV: ≈₹7.97 Billion

10. State-wise Performance Summary

The table below consolidates order volume, GMV, success rate, CSAT, and seller-productivity metrics for all eight states, providing a single reference view for regional comparison.

State	Orders	GMV (₹ Cr)	Success Rate	CSAT	Active Sellers	Orders/Seller
Maharashtra	44,02,171	797.48	97.11%	4.5/5	2,26,990	19.39
Uttar Pradesh	30,85,662	599.27	96.81%	4.54/5	1,65,273	18.67
Karnataka	22,92,793	378.08	96.53%	4.57/5	1,14,404	20.04
Delhi NCR	18,11,617	310.05	96.97%	4.58/5	91,265	19.85
Tamil Nadu	8,50,060	146.58	97.27%	4.56/5	39,236	21.67
Gujarat	6,56,746	127.4	96.81%	4.57/5	32,609	20.14
West Bengal	3,53,334	61.41	96.92%	4.48/5	18,286	19.32
Rajasthan	78,239	13.38	97.14%	4.55/5	5,366	14.58

INSIGHTS:

- Maharashtra leads on both orders and GMV, but Tamil Nadu has the highest orders-per-seller ratio, indicating stronger seller productivity relative to its market size.
- Rajasthan has the lowest orders-per-seller ratio among the eight states, which may reflect an emerging or under-penetrated market.
- Success rate and CSAT stay within a narrow band across states, suggesting operational quality is broadly consistent regardless of state-level demand size.
- This table is a useful starting point for prioritizing seller-acquisition and logistics investment by state.

11. Regional Demand

Purchase demand is distributed across 8 states. Maharashtra has the highest order volume in the sampled data at 44,02,171 orders. Regional demand analysis can support seller coverage, operational planning, service monitoring, and state-level business decisions.

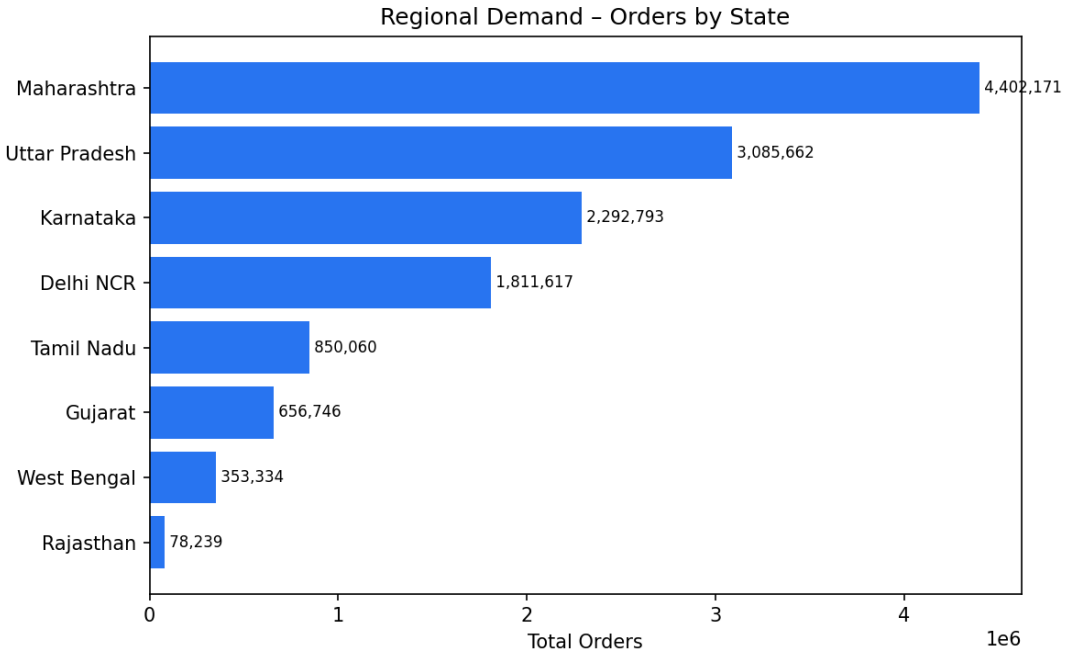


Figure 7. Regional demand – orders by state

INSIGHTS:

- Regional demand can support seller-coverage planning.
- It can also help with operational planning and service monitoring.
- State-level demand information can support regional business decisions.

KPI:

Top-performing state: Maharashtra

Order volume: 4.40 Million

States covered: 8

12. Success Rate

The average order success rate across the sampled records is 96.94%.



Figure 8. Average order success rate

INSIGHTS:

- State-level differences: a state-wise breakdown can reveal which states have relatively higher or lower success rates; states below the overall average may need closer investigation.
- Success rate should be monitored continuously because changes may indicate operational or payment-gateway issues.
- The average CSAT score is 4.54/5. Success rate is an operational reliability metric, whereas CSAT reflects the customer experience captured in the dataset.

KPI:

Average order success rate: 96.94%

Operational status: High order reliability

Categories tracked: 6

13. Active Sellers vs Order Volume

The relationship between active sellers and order volume provides an additional descriptive view of state-level activity. States with more active sellers may generate higher order volumes, but the scatter plot should be interpreted as an association rather than proof of causation.

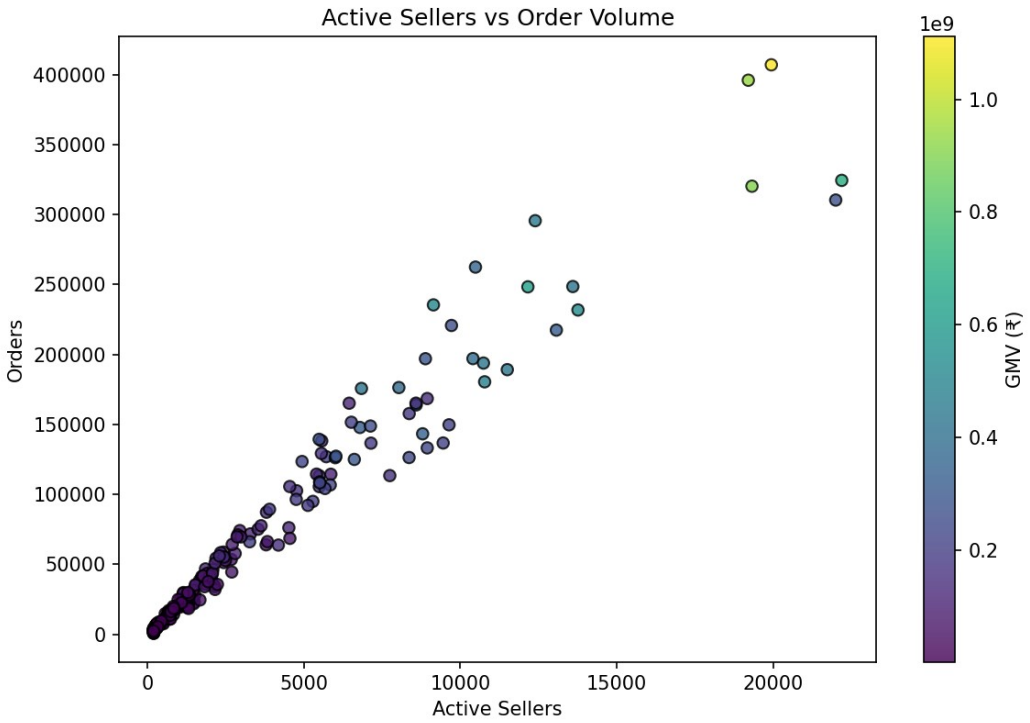


Figure 9. Active sellers vs order volume

INSIGHTS:

- A positive association may indicate that seller coverage is related to order activity.
- However, the chart does not prove that more sellers cause higher order volume.
- Other factors — such as regional demand, logistics reach, and category mix — can also influence order volume.
- This analysis can support seller productivity and coverage planning.

KPI:

Top order-volume state: Maharashtra

Active sellers in that state: ≈227.0K

Order volume: ≈4.40 Million

Relationship: Higher seller activity is associated with higher order volume in the sampled data

14. Weekday vs Weekend Purchase Pattern

Across the row-level observations, average orders are 55,807 on weekdays versus 75,885 on weekends.

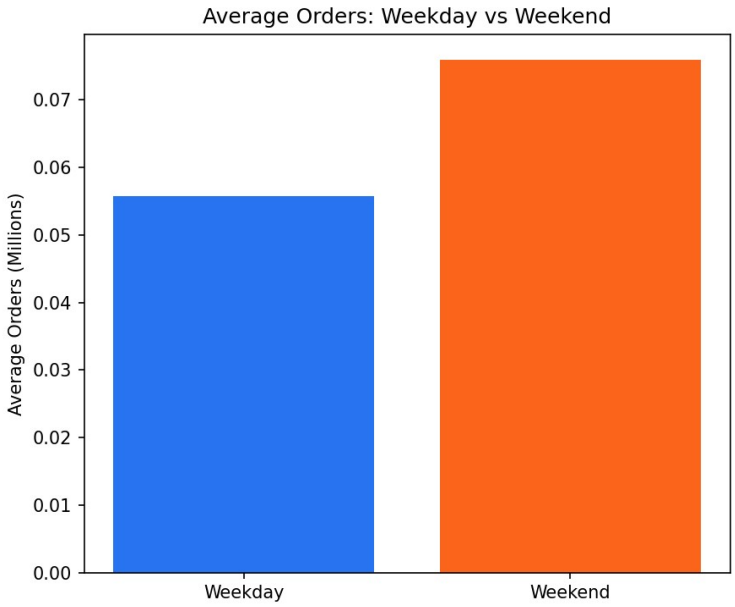


Figure 10. Weekday vs weekend average orders

INSIGHTS:

- Weekend order volumes are higher on average than weekday volumes in the sampled data, consistent with typical leisure-driven e-commerce browsing and buying behavior.
- The dataset is not a complete continuous daily series for every state/category combination.
- Therefore, the result should be treated as a descriptive comparison of the available observations, not a definitive seasonal pattern.

KPI:

Highest orders-per-seller state: Tamil Nadu

Orders per seller: ≈21.67

Lowest orders-per-seller state: Rajasthan

Orders per seller: ≈14.58

15. Customer Satisfaction (CSAT)

The overall row-level average CSAT is 4.54/5. Delhi NCR has the highest average CSAT at 4.58, while West Bengal has the lowest at 4.48. CSAT is useful as a customer-experience KPI alongside operational measures such as success rate.

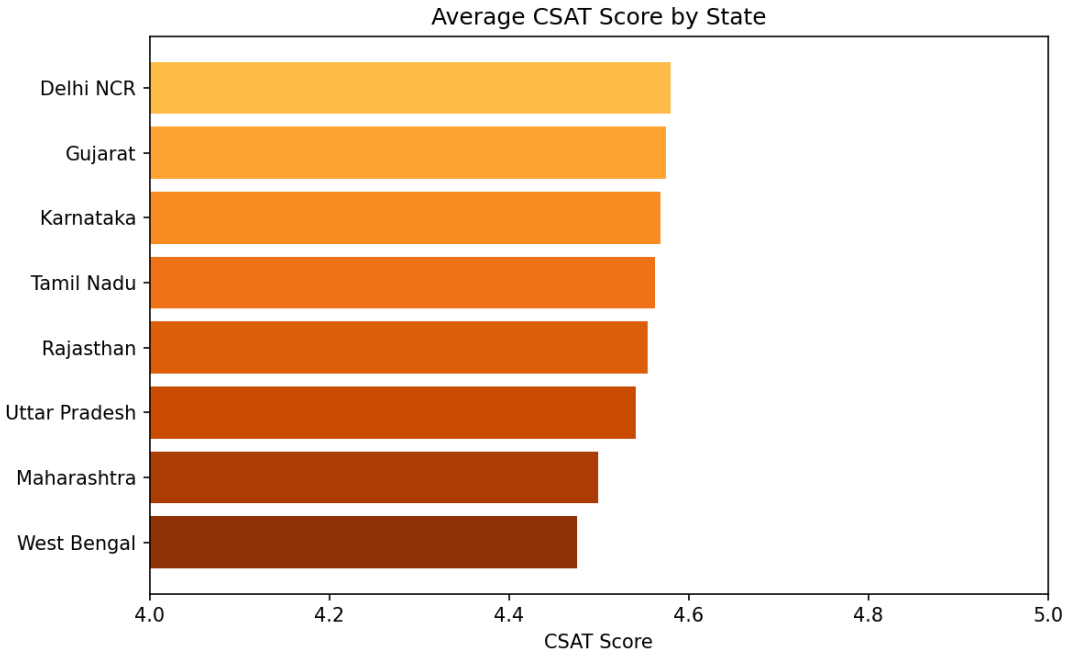


Figure 11. Average CSAT score by state

INSIGHTS:

- This shows some variation in customer satisfaction between states.
- Delhi NCR represents the strongest CSAT performance among the states shown.
- West Bengal may require closer investigation to understand its comparatively lower score.
- CSAT should be considered alongside operational KPIs such as success rate.

KPI:

Highest CSAT state: Delhi NCR

CSAT: 4.58 / 5

Lowest CSAT state: West Bengal

CSAT: 4.48 / 5

16. Limitations & Assumptions

Every finding in this report should be read against the boundaries of the underlying data. The points below summarize what this analysis can and cannot support.

Scope limitations

- The dataset is a 200-record sample across only 5 dates — it is not a continuous daily series and does not capture month-end, festive-sale, or seasonal effects.
- Records are pre-aggregated at the date/state/category level; individual customer- or order-level behavior cannot be examined.
- The dataset was synthetically generated to resemble plausible Flipkart-style order patterns; it does not reflect actual company-reported figures and should not be cited as real Flipkart performance data.
- Zero downtime-flagged records in the sample should not be read as "the platform never experiences downtime" — it only reflects the absence of flagged incidents in this specific sample.

Interpretation caveats

- Associations shown in scatter plots (e.g., active sellers vs. order volume) describe correlation, not causation.
- Category and state rankings apply only within the sampled window and may shift with a longer or differently sampled period.
- Success rate and CSAT are averaged across records; they can mask meaningful variation at finer time or category granularity.

Summary of Key Metrics

Metric	Value
Total orders	1,35,30,622
Total GMV	₹2,433.65 crore
Average success rate	96.94%
Average CSAT score	4.54 / 5
States tracked	8
Purchase categories tracked	6
Dataset records	200
Downtime-flagged records	0

Next Steps to Move from Descriptive → Predictive

- Demand forecasting by state and purchase category to estimate future order volumes.
- Anomaly detection for sudden changes in success rate, GMV, or order volume.
- Downtime-impact analysis linking operational flags to order failures and lost GMV.
- Seller productivity analysis using orders per active seller.
- Trend and cohort analysis when longer historical data becomes available.

This report answers what happened in the sampled dataset. The next analytical layer would use a longer time period and richer event-level data to explain why changes occurred and to forecast what is likely to happen next.

Glossary

I. GMV — Gross Merchandise Value: Total monetary value of orders processed/sold through a platform before deducting fees, discounts, refunds, or other adjustments. The exact definition can vary by dataset/platform.

II. GST — Goods and Services Tax: India's unified indirect tax applied on the sale of goods and services, including most e-commerce transactions.

III. ONDC — Open Network for Digital Commerce: An open network/protocol for digital commerce in India. Marketplaces such as Flipkart can interoperate with or run seller listings alongside ONDC, but ONDC itself is not a government-owned e-commerce platform.

IV. CSAT — Customer Satisfaction Score: A customer-experience metric usually collected through post-purchase or post-delivery surveys, often asking customers to rate their satisfaction.

V. Active Sellers: The count of marketplace sellers with at least one fulfilled order within the reporting window.